

Comparison of gliding box and box-counting methods in soil image analysis

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Abstract

Three soil images with different spatial arrangements and porosities ranging from 5% to 47% were analyzed to calculate their generalized dimensions (D_q). The first partitioning method applied in these calculations was box-counting, in which a grid is used to study the local distribution of porosity at different scales. In all the images, porosity was found to exhibit multi-scaling behavior in the larger box sizes only, i.e., with side lengths ranging from 64 to 256 pixels. Estimates of D_q were obtained by restricting multifractal analysis (MFA) to these box sizes, but this resulted in very few points for linear regression analysis and only a small number of boxes per size due to image size limitations. The gliding box method was subsequently applied to the same range of box sizes using the multiplier method. This yielded less uncertain D_q values, particularly for negative values of q . The gliding box method was therefore found to be more suitable for MFA where the range of usable scales is narrow. Both the numerical differences between D_q values and their standard errors are discussed.

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1. Introduction

Although many parameters may be used to describe irregular morphology, the spatial arrangement of the most prominent features is a challenging problem across a wide range of disciplines (Ripley, 1988; Griffith, 1988; Baveye and Boast, 1998). In the case of 2-dimensional images of soil thin sections, fractal techniques have been used in a number of endeavors to describe the spatial structure (Brakensiek et al., 1992; Peyton et al., 1994; Anderson et al., 1996; Pachepsky et al., 1996; Giménez et al., 1997, 1998; Oleschko et al., 1997, 1998; Oleschko, 1998; Bartoli et al., 1999; Dathe et al., 2001).

Interest has recently turned to the multifractal analysis of images of porous media, a technique that

has been widely used for pore systems in rocks (Muller and McCauley, 1992; Saucier, 1992; Muller et al., 1995; Muller, 1996; Saucier and Muller, 1999). Applications in connection with soils, such as the study by Posadas et al. (2003), have been reviewed by Tarquis et al. (2003). A geometric multifractal (Tel and Vicsek, 1987) is a non-uniform fractal which, unlike uniform fractals, exhibits local fluctuations in density. Consequently, it cannot be characterized by a single dimension but calls, rather, for a sequence of generalized fractal dimensions. Multifractal analysis may prove useful for extracting these dimensions from soil images which contain complex spatial distributions of solids and/or pores, and where prominent local variations in density or porosity are observed.

Several authors maintain that the exact value of the generalized dimension cannot be readily calculated (Crawford et al., 1999; Bird et al., in press). Vicsek

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(1990), for instance, proposed practical methods to compute the generalized dimension, showing that the standard methods for determining fractal dimensions must be applied with some caution.

The main difficulty in using multifractal procedures is that the ideal limit cannot be reached in practice (Buczkowski et al., 1998). As a general rule, moreover, only a limited number of points on the spectrum are used to describe the major properties of the associated measure. In addition, multifractal analysis (MFA) involves partitioning the space studied into boxes to construct multiple-scale samples. The number of samples for a given scale is restricted by the size of the space partitioned and the resolution of the data. As a result, these factors often influence the quality of statistical estimation in MFA (Cheng and Agerberg, 1996).

The aim of the present study is to examine the effect of the box size range chosen and the subdivision method applied in the calculation of generalized fractal dimensions (D_q). To this end, three soil images with different spatial arrangements were analyzed to obtain their D_q values.

2. Materials and methods

2.1. Images

Three soil samples, kindly furnished by the University of Guelph, Department of Land Resource Science, were selected for their different void size distributions and connectivity. Soil porosity ranged from 5% to 47% in these samples, which were prepared for image analysis as described by Protz and VandenBygaart (1998).

Data were obtained by imaging the thin sections with a Kodak 460 RGB camera using transmitted and circularly polarized illumination. The images were cropped from 3060×2036 pixels to 3000×2000 pixels. A pixel size of $18.6 \times 18.6 \mu\text{m}$ was used (see Protz and VandenBygaart, 1998, for further details.). The data were classified and the void bitmap separated with EASI/PACE software. The images for the three soils are shown in Fig. 1. Characteristic soil texture, organic matter content, sample depth, type of horizon and soil classification are given in Table 1.

An area of 1024×1024 pixels at the upper left corner of the original images (outlined by a solid line in the images in Fig. 1) was selected for box-counting analysis to avoid possible edge effects in box-counting calculations. The gliding box-method was applied to the same area to ensure comparability.

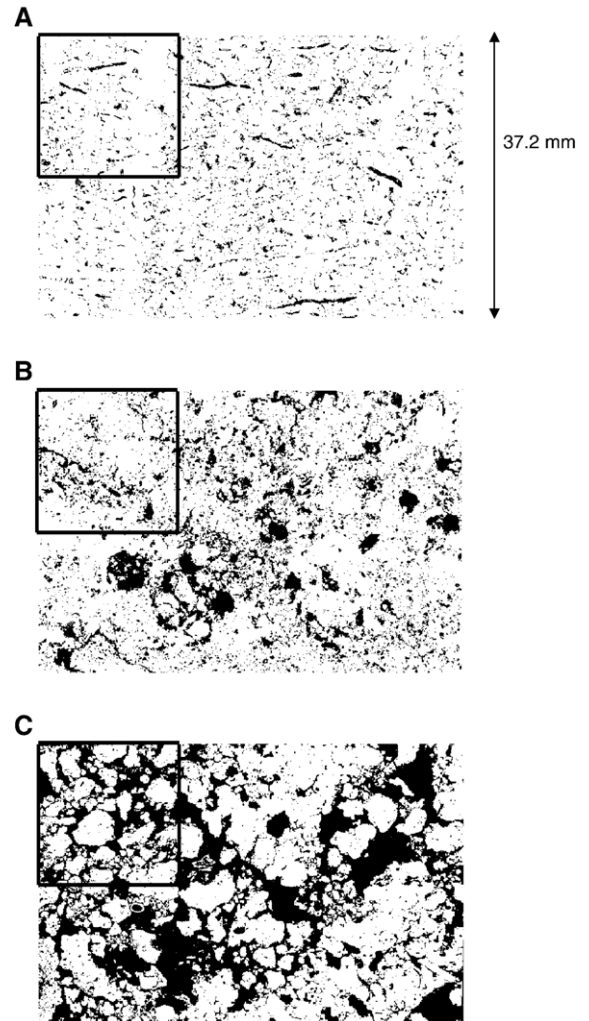


Fig. 1. Soil binary images, pore phase in black pixels, for: (A) ADS, (B) BUSO, (C) EHV1, with porosities of 5.65%, 19.17% and 46.67%, respectively. The selected area, outlined by a solid line, contains 1024×1024 of the total 2000×3000 pixels in the image.

2.2. Calculation of generalized fractal dimensions

MFA in 2-D images involves partitioning the plane into boxes to construct samples with multiple scales. The box-counting (BC) method combines pixels to form larger, mutually exclusive boxes each containing a different set of pixels. Given an $L \times L$ -pixel image partitioned to a box size of $\delta \times \delta$, the number of boxes will follow the proportion of line size $\delta(n(\delta))$ is:

$$n(\delta) \propto \left(\frac{L}{\delta}\right)^2 \quad (1)$$

If δ is overly large, the resulting number of samples may not be sufficient for statistical analysis.

Table 1

Soil classification, porosity and selected properties of the three soils

Image code	Soil classification ^a	Horizon		Porosity (%)		Properties			
		Type	Depth (cm)	Total image	Selected image area	Organic matter (%)	Sand (%)	Silt (%)	Clay (%)
ADS	Entisol	BC	20–32	6.40	5.25	1.3	2.7	63.5	33.8
BUSO	Spodosol	C	65–73	18.38	19.17	0.1	54.0	15.0	31.0
EHV1	Histosol	H/Ah	0–1	38.28	46.67	33.2	38.0	30.0	32.0

^a According to the U.S. soil taxonomy.

The gliding box (GB) method was originally used in lacunarity analysis (Allain and Cloître, 1991) and later modified by Cheng (1997a,b) for application to MFA. The GB method essentially constructs samples by gliding a box of a certain line size (δ) over the grid map in all possible directions. An “up-scaling” partitioning process begins with a minimum line size box ($\delta_{\min}$), which is steadily enlarged to a specific size smaller than L . The proportionality of the number of boxes of linear size δ ($n^*(\delta)$) is:

$$n^*(\delta) \propto \left(\frac{L - \delta + \delta_{\min}}{\delta_{\min}} \right)^2 \text{ or } n^*(\delta) \propto \left(\frac{L}{\delta_{\min}} - \frac{\delta}{\delta_{\min}} + 1 \right)^2 \quad (2)$$

where $\delta_{\min} \leq \delta \leq L$. This relationship is often expressed as:

$$n^*(\delta) \propto \left(\frac{L}{\delta_{\min}} - \varepsilon + 1 \right)^2 \quad (3)$$

where ε is the ratio between the box line size and the minimum size chosen ($\delta/\delta_{\min}$).

One advantage to the GB method is that the larger sample size it provides usually yields better statistical results. Because this partitioning overlaps, the measures defined by these boxes are not statistically independent and the definition of the measure in gliding boxes differs, as explained below.

2.2.1. Box-counting method

The generalized dimension calculated using the BC technique essentially reflects the mass contained in each box. An image is divided into $n(\delta)$ boxes, and the fraction of pore space (μ_i) in each box is calculated from:

$$\mu_i = \frac{m_i}{m_T} = \frac{m_i}{\sum_{i=1}^{n(\delta)} m_i} \quad (4)$$

where m_i is the number of pore class pixels in box i and m_T is the total number of pore class pixels in an image. In this case, the pore space area is the measure and the

square grid the support. The generating function ($\chi(q, \delta)$) is then defined, as:

$$\chi(1, \delta) = \sum_{i=1}^{n(\delta)} [\chi_i(q, \delta)] \quad q \in \Re \quad (5)$$

where

$$\chi_i(q, \delta) = \mu_i^q = \left(\frac{m_i}{\sum_{i=1}^{n(\delta)} m_i} \right)^q \quad (6)$$

χ_i is a weighted measure that represents the percentage of pore space in box i , q is the weight or moment of the measure, and $\Re$ is the set of real numbers. When computing the number of boxes of size δ , the possible values of m_i range from 0 to $\delta \times \delta$. So if $N_j(\delta)$ is the number of boxes containing j pixels of pore space in a given grid, Eqs. (5) and (6) can then be written (Barnsley et al., 1988) as follows:

$$\chi(q, \delta) = \sum_{i=1}^{n(\delta)} (\mu_i)^q = \sum_{i=1}^{n(\delta)} \left(\frac{m_i}{m_T} \right)^q = \sum_{j=1}^{\delta^2} N_j(\delta) \left(\frac{j}{m_T} \right)^q \quad (7)$$

where: $m_T = \sum_{j=1}^{\delta^2} j N_j(\delta)$

Using the distribution function $N_j(\delta)$ simplifies calculations and reduces computational errors (Barnsley et al., 1988).

A log–log plot of a self-similar measure, $\chi(q, \delta)$, vs. δ at various values for q gives:

$$\chi(q, \delta) \sim \delta^{\tau(q)} \quad (8)$$

where $\tau(q)$ is the mass exponent for q (Hentschel and Procaccia, 1983; Feder, 1989). $\tau(q)$ can be written as:

$$\tau(q) = \lim_{\delta \rightarrow 0} \frac{\log(\chi(q, \delta))}{\log(\delta)} \quad (9)$$

The generalized dimension, D_q , can then be introduced through the following scaling relationship (Hentschel and Procaccia, 1983; Feder, 1989):

$$D_q = \lim_{\delta \rightarrow 0} \frac{\log[\chi(q, \delta)]}{(q-1)\log\delta} \quad (10)$$

And, therefore

$$\tau(q) = (q-1)D_q \quad (11)$$

The following relationship is used where $q=1$, since Eq. (11) is not applicable in such cases:

$$D_1 = \lim_{\delta \rightarrow 0} \frac{\sum_{i=1}^{n(\delta)} \chi_i(1, \delta) \log[\chi_i(1, \delta)]}{\log \delta} \quad (12)$$

The physical interpretation of the system is only clearly reliable where $q=0$, $q=1$ or $q=2$. D_1 , for

instance, has been related to the permeability of sedimentary rocks in several papers (Muller and McCauley, 1992; Muller et al., 1995; Muller, 1996). The generalized dimensions, D_q , for $q=0$, $q=1$ and $q=2$ are known as the capacity, information, and correlation dimensions, respectively (Hentschel and Procaccia, 1983). The capacity dimension is the box-counting or fractal dimension of the support. The information dimension is related to system entropy, whereas the correlation dimension computes the correlation of the measures contained in boxes of various sizes (Posadas et al., 2003).

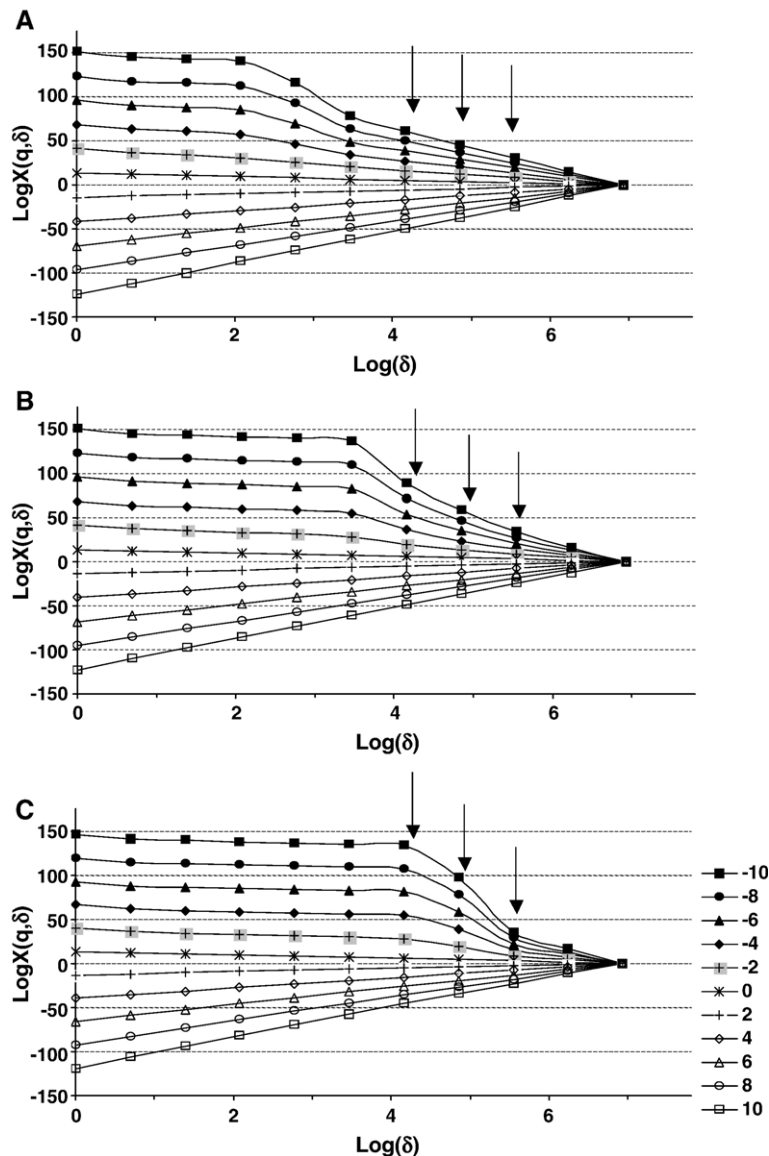


Fig. 2. Bi-log plot of $\chi(q, \delta)$ versus box size (δ) at different values of mass exponent (q): (A): ADS; (B) BUSO, (C) EVH1. The arrows indicate the three selected points for linear regression.

2.2.2. Gliding box method

As in the BC method, a definition of the mass contained in each box (number of pore pixels) must be given in the gliding box procedure. Let $n^*(m, \delta)$ be the number of gliding boxes of size δ and mass m . Dividing by $n^*(\delta)$ yields the probability function, $\beta(m, \delta)$, for a gliding box of size δ and mass m .

The statistical moment of this distribution is:

$$\chi^*(q, \delta) = \sum_{m=1}^{\delta \times \delta} \beta(m, \delta) m^q = \frac{1}{n^*(\delta)} \sum_{i=1}^{n^*(\delta)} m_i^q(\delta) \approx E\{m^q\} \quad (13)$$

where $\chi^*(q, \delta)$ is the moment of order q of $\beta(m, \delta)$, $E\{\}$ is the mathematical expectation and the sum includes all gliding boxes with $m_i > 0$.

The difference between $\chi^*(q, \delta)$ (Eq. (13)) and $\chi(q, \delta)$ (Eq. (7)) is that:

$$\chi(q, \delta) \approx \left(\frac{L}{\delta}\right)^2 E\{m^q\} \quad (14)$$

Combining Eqs. (13) and (14) yields:

$$\chi(q, \delta) \approx \left(\frac{L}{\delta}\right)^2 \chi^*(q, \delta) \quad (15)$$

Substituting this relation in Eqs. (8) and (9) gives:

$$\tau(q) = \lim_{\delta \rightarrow 0} \frac{\log(\chi^*(q, \delta))}{\log(\delta)} + \lim_{\delta \rightarrow 0} \frac{2 \log\left(\frac{L}{\delta}\right)}{\log(\delta)} \quad (16)$$

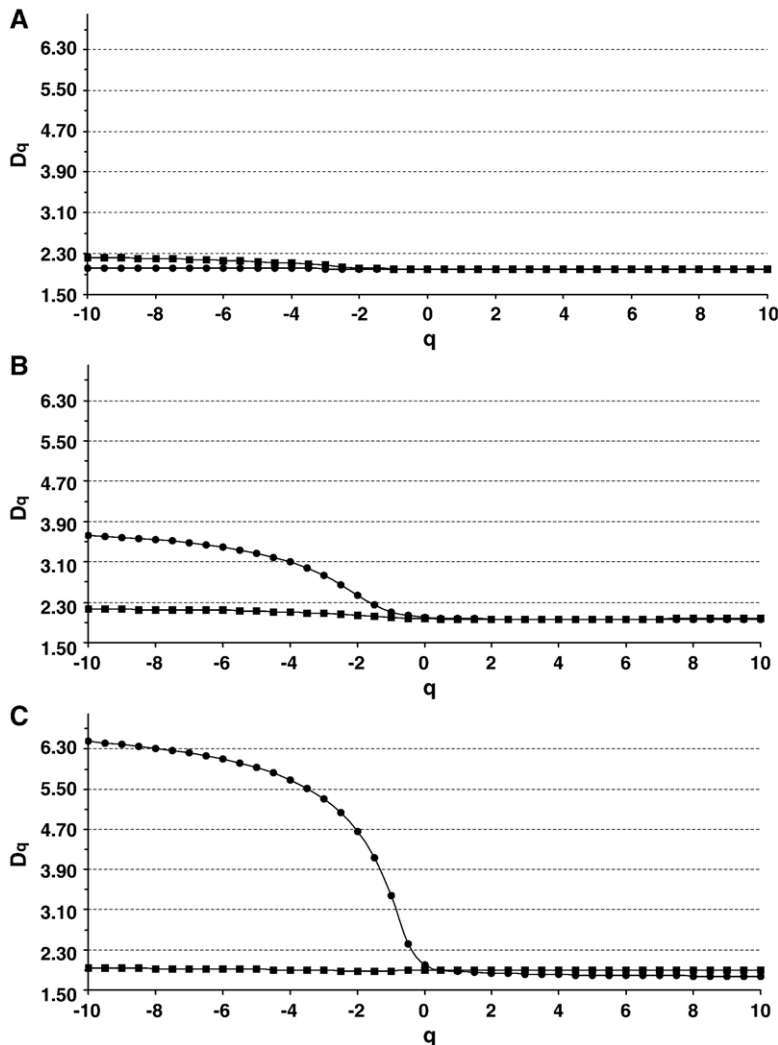


Fig. 3. Generalized dimensions (D_q) from $q = -10$ to $q = +10$ for all points on the regression line (solid squares) and for three selected points based on the bi-log plot of $\chi(q, \delta)$ (solid circles) for each image: (A) ADS; (B) BUSO, (C) EVH1.

The second limit is readily solved, giving a value of -2 . It is concluded that (Cheng, 1997a,b):

$$\tau(q) = \lim_{\delta \rightarrow 0} \frac{\log(\chi^*(q, \delta))}{\log(\delta)} - 2 \quad (17)$$

where 2 is the value of the dimension of the Euclidean plane containing the image.

In the present study the multiplier method, based on Eq. (18) below, was applied to estimate $\tau(q)$ (see Cheng, 1999 for further details):

$$\langle \tau(q) \rangle + 2 = - \frac{\log(\langle M(q, \delta) \rangle)}{\log(\varepsilon)} \quad (18)$$

where $\langle \rangle$ stands for the first statistical moment and M represents the multiplier measured on each pixel as:

$$M(q, \delta) = \left(\frac{\mu(\delta_{\min})}{\mu(\delta)} \right)^q \quad (19)$$

The advantage of using Eq. (18) compared to Eq. (9) is that, since the estimation is independent of box size δ , successive box sizes may be used to estimate $\tau(q)$. A condition to applying the gliding box analysis, however, is that $\mu(\delta_{\min})$ may not be nil, for otherwise the expression $\left(\frac{\mu(\delta_{\min})}{\mu(\delta)} \right)^q$ is always zero or undefined for all values of δ .

The resulting estimate of $\langle \tau(q) \rangle$ may be applied in Eq. (11) to estimate D_q . Following Saucier and Muller (1999), Eq. (20) is used to estimate the information dimension when $q=1$:

$$D_1 = 2D_2 - D_3. \quad (20)$$

3. Results and discussion

3.1. The generating function with the box-counting method

After calculating $\chi(q, \delta)$ for the three binary images, $\log \chi(q, \delta)$ was plotted against $\log(\delta)$. A clear pattern can be discerned in all of the plots. In Fig. 2, for example, where q is negative, two regions can be distinguished: one where the relationship between $\log(\delta)$ and $\log(\chi(q, \delta))$ is linear and the other in which the value of $\log(\chi(q, \delta))$ is almost constant at any value of $\log(\delta)$. The box size at which the behavior differs is around 64 pixels in all three images. For positive values of q , these two phases are not visible (see Fig. 2).

The existence of a plateau phase for $\log(\chi(q, \delta))$ can be explained by the nature of the measure studied. At δ values close to unity, the variation in the number of black pixels is based on only a few pixels, with a minimum for $\delta=1$, where the measure may assume only two values, 0 or 1.

Table 2

Results obtained by box-counting and gliding box methods for ADS (box size from 64 to 256 pixels)

q	Gliding box method ($n=192$)				Box-counting method ($n=3$)			
	$\tau(q)$	S.E.	r	t -value*	$\tau(q)$	S.E.	r	t -value**
-10	-22.23	0.004	-0.999	4758	-22.27	0.112	-0.999	199
-9	-20.17	0.004	-0.999	4980	-20.20	0.081	-0.999	251
-8	-18.13	0.003	-0.999	5231	-18.14	0.057	-0.999	321
-7	-16.10	0.003	-0.999	5516	-16.10	0.039	-0.999	417
-6	-14.07	0.002	-0.999	5840	-14.07	0.025	-0.999	553
-5	-12.05	0.002	-0.999	6200	-12.05	0.016	-0.999	751
-4	-10.03	0.001	-0.999	6622	-10.03	0.009	-1.000	1059
-3	-8.02	0.001	-0.999	7093	-8.02	0.005	-1.000	1584
-2	-6.01	0.001	-0.999	7626	-6.01	0.002	-1.000	2650
-1	-4.00	0.001	-0.999	8235	-4.00	0.001	-1.000	5880
0	-2.00	0.000	-1.000	—	-2.00	0.000	1.000	—
1	+2.00	0.001	+0.999	9729	+2.00	0.001	+1.000	7115
2	+2.00	0.001	+0.999	10,666	+2.00	0.001	+1.000	3882
3	+4.00	0.001	+0.999	11,761	+3.99	0.001	+1.000	2811
4	+5.99	0.001	+0.999	13,065	+5.99	0.003	+1.000	2278
5	+7.99	0.001	+0.999	14,630	+7.98	0.004	+1.000	1962
6	+9.98	0.001	+1.000	16,527	+9.98	0.006	+1.000	1752
7	+11.98	0.001	+1.000	18,829	+11.97	0.007	+1.000	1604
8	+13.97	0.001	+1.000	21,585	+13.96	0.009	+1.000	1494
9	+15.96	0.001	+1.000	24,722	+15.95	0.011	+1.000	1409
10	+17.95	0.001	+1.000	27,838	+17.94	0.013	+1.000	1343

Critical values: * $t_{0.05,190}=2.346$, ** $t_{0.05,1}=63.657$. Dashed lines: t -value > 10000.

Table 3

Results obtained by box-counting and gliding box methods for BUSO (box size from 64 to 256 pixels)

q	Gliding box method ($n=192$)				Box-counting method ($n=3$)			
	$\tau(q)$	S.E.	r	t -value*	$\tau(q)$	S.E.	r	t -value**
-10	-37.50	0.453	-0.923	78	-39.92	3.207	-0.997	12
-9	-33.89	0.399	-0.981	80	-35.90	2.909	-0.997	12
-8	-30.26	0.345	-0.982	82	-31.87	2.621	-0.997	12
-7	-26.62	0.290	-0.983	85	-27.83	2.346	-0.996	12
-6	-22.97	0.233	-0.985	90	-23.75	2.087	-0.996	11
-5	-19.27	0.176	-0.988	98	-19.64	1.838	-0.996	11
-4	-15.51	0.116	-0.991	116	-15.50	1.562	-0.995	10
-3	-11.59	0.058	-0.996	166	-11.34	1.153	-0.995	10
-2	-7.45	0.012	-0.999	449	-7.34	0.522	-0.997	14
-1	-4.21	0.001	-0.999	1900	-4.21	0.072	-0.999	59
0	-2.00	0.001	-1.000	900	-2.00	0.000	-1.000	–
1	+1.99	0.000	+0.999	1000	+1.98	0.003	-0.999	502
2	+1.92	0.001	+0.999	3000	+1.97	0.004	0.999	534
3	+3.79	0.004	+0.999	1490	+3.93	0.006	0.999	682
4	+5.62	0.011	+0.999	724	+5.89	0.006	0.999	954
5	+7.36	0.022	+0.999	409	+7.85	0.001	1.000	1302
6	+9.03	0.042	+0.998	263	+9.80	0.003	1.000	2838
7	+10.62	0.066	+0.997	190	+11.76	0.000	1.000	–
8	+12.16	0.094	+0.995	150	+13.72	0.002	1.000	1500
9	+13.68	0.123	+0.993	130	+15.68	0.006	1.000	2600
10	+15.18	0.152	+0.991	113	+17.64	0.010	1.000	1300

Critical values: $*t_{0.05,190}=2.346$, $**t_{0.05,1}=63.657$. Dashed lines: t -value > 10000.

Thus, where δ is small, proportionally speaking, relationships among the values remain approximately constant. After a certain critical box size, however, a scaling pattern

appears. This could be interpreted as the representative elementary area needed to study this scaling property (VandenBygaart and Protz, 1999). Similar behavior has

Table 4

Results obtained by box-counting and gliding box methods for EHV1 (box size from 64 to 256 pixels)

q	Gliding box method ($n=192$)				Box-counting method ($n=3$)			
	$\tau(q)$	S.E.	r	t -value*	$\tau(q)$	S.E.	r	t -value**
-10	-39.00	0.983	-0.923	38	-70.98	10.95	-0.988	6
-9	-35.38	0.881	-0.924	38	-63.92	9.767	-0.989	7
-8	-31.77	0.777	-0.904	38	-56.85	8.571	-0.988	7
-7	-28.16	0.674	-0.927	39	-49.78	7.372	-0.989	7
-6	-24.57	0.571	-0.930	40	-42.69	6.161	-0.990	7
-5	-21.00	0.468	-0.933	41	-35.57	4.936	-0.991	7
-4	-17.44	0.366	-0.938	42	-28.43	3.681	-0.992	8
-3	-13.86	0.262	-0.945	45	-21.23	2.386	-0.994	9
-2	-10.12	0.155	-0.958	53	-13.98	1.035	-0.997	13
-1	-6.46	0.181	-0.981	34	-6.76	0.144	-0.999	47
0	-2.00	0.001	-0.999	200	-1.99	0.005	-0.999	413
1	+1.90	0.003	+0.999	864	+1.87	0.008	-0.999	222
2	+1.62	0.023	+0.999	156	+1.84	0.014	0.999	132
3	+2.76	0.115	+0.999	120	+3.64	0.034	0.999	106
4	+4.36	0.112	+0.992	57	+5.42	0.061	0.999	88
5	+5.09	0.120	+0.989	40	+7.19	0.095	0.999	75
6	+6.38	0.240	+0.981	35	+8.95	0.137	0.999	66
7	+7.19	0.211	+0.978	34	+10.71	0.184	0.999	58
8	+8.08	0.375	+0.969	27	+12.47	0.238	0.999	52
9	+9.09	0.434	+0.909	25	+14.22	0.296	0.999	48
10	+09.69	0.512	+0.899	23	+15.97	0.359	0.999	44

Critical values: $*t_{0.05,190}=2.346$, $**t_{0.05,1}=63.657$. Dashed lines: t -value > 10,000.

been reported in other binary images, such as dye patterns in soil (Tarquis et al., in press).

3.2. Generalized dimensions using the box-counting method

The D_q values, particularly for $q < 0$, varied considerably depending on whether all the regression points or only the points in the linear region were used in the analysis (Fig. 3). It may be deduced from the figure that in between these two approaches, any D_q value is possible, and will depend on the number of regression points chosen for the analysis, but where $q \geq 0$ the differences are not significant. This

is a recurrent observation ever since multifractal analysis was first applied to experimental results and a careful study of correlation coefficients or R^2 values is necessary (Vicsek, 1990; Evertsz and Mandelbrot, 1992; Crawford et al., 1999; Cheng, 1999; Posadas et al., 2003; Bird et al., in press; Tarquis et al., in press). The inference to be drawn from the—in this case dramatic—changes in D_q values is that the amplitude of the dimensions ($D_{-10} - D_{+10}$) as used in several studies (see Tarquis et al., 2003 and references there in) cannot be compared or calculated.

The differences in the D_q plots (Fig. 3, solid circles) are observed chiefly for negative values of q . In

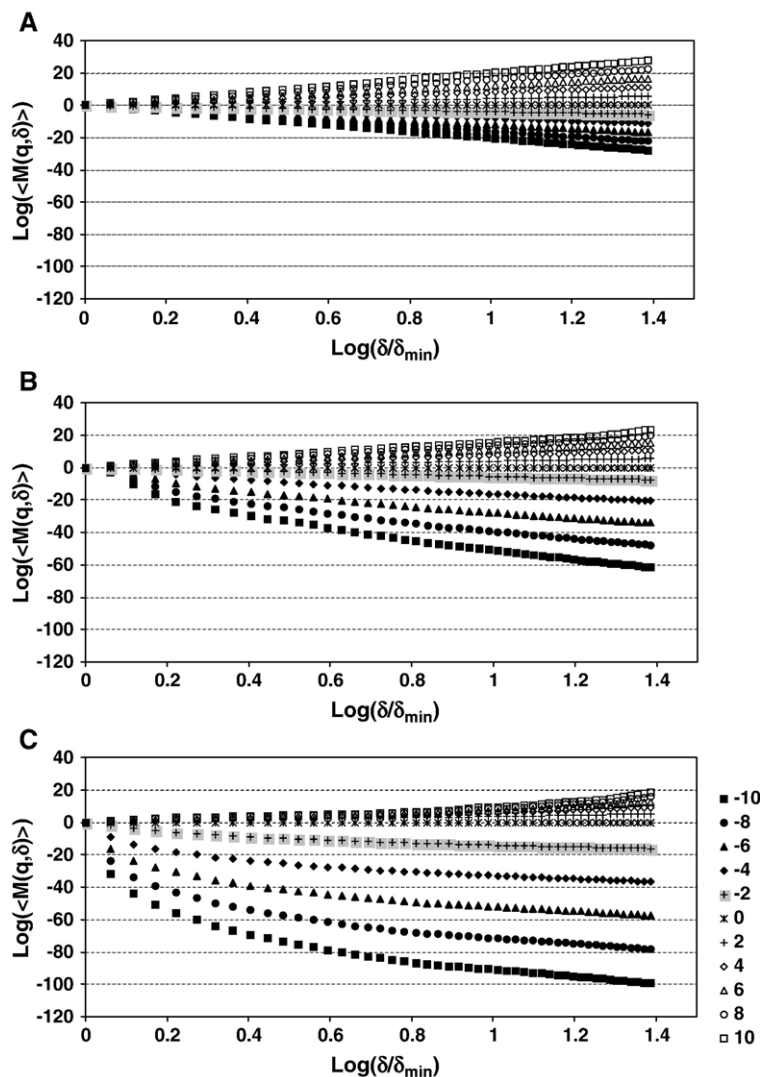


Fig. 4. Bi-log plot of $\langle M(q, \delta) \rangle$ versus box size ratio ($\delta/\delta_{\min}$) at different values of mass exponent (q): A) ADS; B) BUSO, C) EVH1. The $\delta_{\min}$ value used was 64 pixels.

particular, ADS (Fig. 3A, solid circles)—a silty, low porosity soil (Table 1)—quite obviously does not exhibit multifractal behavior since the D_q values are approximately constant.

A value of 2 (plane dimension) was found for all D_0 ; this overestimation is due to the size range selected to obtain an optimum fit for the linear regressions for each q value.

With the aim to test the significance of the slope ($\tau(q)$) the two-tailed Student's t was calculated. The values for $\tau(q)$, standard errors (S.E.), correlation coefficients (r) and Student's t in box sizes ranging from 64 to 256 pixels are shown in Tables 2, 3 and 4 for ADS, BUSO and EHV1, respectively. The very high r values, all above 0.98, observed for this range of box sizes is hardly surprising, given that only three points were used. A comparison of the t -values in the tables with the critical threshold (63.657) shows that for ADS the $\tau(q)$ values are statistically significant from $q=-10$ to $q=+10$, whereas for BUSO they are significant from $q=0$ to $q=+10$, and for EHV1 from $q=0$ to $q=+6$.

The number of boxes of each size, in turn, was very low: 64 for 128×128 -pixel and 16 for 256×256 -pixel boxes in a 1024×1024 -pixel image. This size limitation can be avoided by using the gliding box method, discussed in the following section.

3.3. Generalized dimensions using the gliding box method

After calculating $\langle M(q, \delta) \rangle$, $\log \langle M(q, \delta) \rangle$ was plotted against $\log \frac{\delta}{\delta_{\min}}$. The relationship found was linear in almost all the plots, as expected when a large number of points is used to calculate a linear regression (Fig. 4). Unlike the case of the other images (Fig. 4A, B), soil EHV1 (Fig. 4C) did not display a linear relationship when $q < -6$.

The $\tau(q)$, standard error, correlation coefficient and Student's t values for box sizes ranging from 64 to 256 pixels for ADS, BUSO and EHV1 are given in Tables 2, 3 and 4, respectively. The lowest values of r , around 0.93, are observed for negative q in EHV1 and for $q=-10$ in BUSO, which is consistent with Fig. 4C and B. A comparison of the t -values in the tables with the critical threshold value (2.346) shows that for all three images the $\tau(q)$ values were statistically significant over the entire range from $q=-10$ to $q=+10$.

In the three cases studied the standard errors found with the gliding box method were consistently smaller than observed in the box-counting method. The average

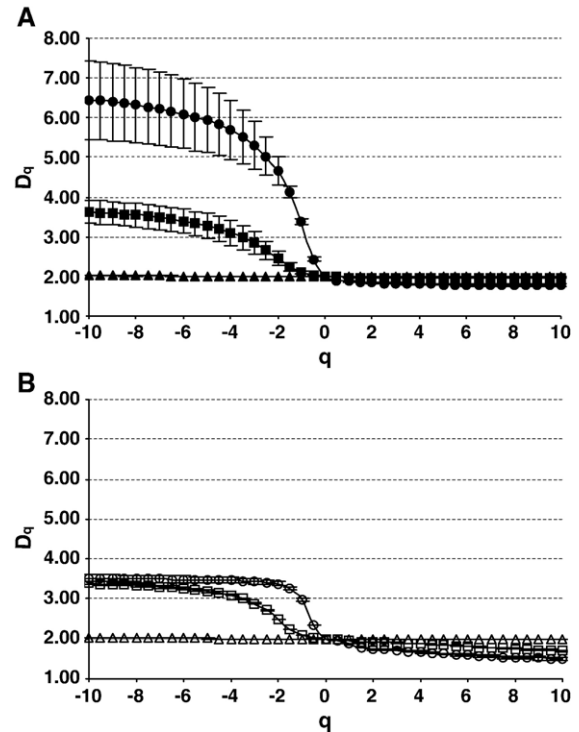


Fig. 5. Generalized dimensions (D_q) from $q=-10$ to $q=+10$ for the three soil thin sections (ADS, triangles; BUSO, squares and EHV1, circles) based on: (A) the box-counting method and (B) gliding box method. The box sizes used were from 64 pixels to 256 pixels. The S. E. of D_q values are represented by line bars.

t -test significance levels of the results obtained using the gliding box method were higher than for the box-counting method, especially for negative values of q .

Curves comparing the three soil images as analyzed by the two methods are plotted in Fig. 5. The standard errors for the D_q values were, logically, lower in the GB than in the BC method (see bars in Fig. 5). These errors were based on the error estimates of $\tau(q)$. The main differences found in the BC method (Fig. 5A) are in the negative q range, whereas in the GB procedure differences were also observed in the positive q range. Fig. 5 also provides a comparison between the D_q values obtained with the two methods. The two methods are similar for soil ADS (solid, Fig. 5A, and empty, Fig. 5B, triangles). The differences between the two methods for soils BUSO and EHV1 are wider for negative values of q , although D_q decay is greater in the GB method for the positive values (Fig. 5B).

4. Conclusions

“Fractal” and “multifractal” are concepts that have grown increasingly popular in recent years in the

analysis of porous materials, soils included, along with the development of fractal models of porous media. The multi-scale heterogeneity of soil structure must be characterized in useful ways for the intents and purposes of modeling, being the generalized dimensions one of them. This implies that D_q values obtained should have as less uncertain as possible and the box size range used should be included in the MFA. Otherwise, this approach may lead to misinterpretations.

By generating a substantially higher number of large size boxes and higher number of points for the linear regression analysis, the “gliding box” method of multi-fractal modeling of soil images yields less uncertain results than does the “box-counting” method.

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